

Section 3: System Design Fundamentals

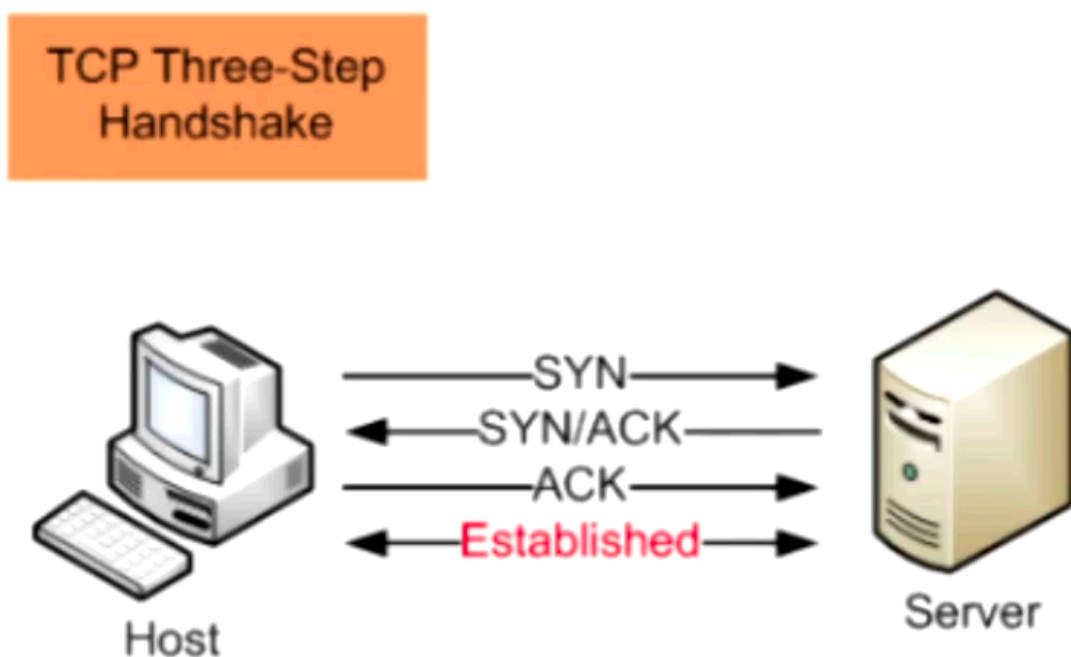
TCP & UDP

TCP and UDP are two of the most fundamental communication protocols in networking and system design. If you have ever wondered how data travel across the internet, how messages are reliably delivered, or why some of your applications work faster than others, then this session is for you. We'll be breaking down the core differences between TCP and UDP, we'll explore their strengths and weaknesses, and understand when to use what.

By the end of this session, you will have a clear understanding of why some applications prioritise reliability, while others focus on speed. This knowledge is crucial for designing scalable, efficient and robust systems. Whether you're preparing for system design interviews, or working on real world distributed applications, this knowledge is going to help you.

What is the Transmission Control Protocol (TCP)?

TCP is one of the widely used protocols for communication over the internet and it plays a critical role in ensuring that data is transmitted accurately and in order. The first thing that we need to know about this is that this is a connection oriented protocol. This means that before any data is exchanged, a connection must be established between the sender and the receiver. Think of it like making a phone call, you don't start talking unless another person picks up and acknowledges the connection.



How does TCP achieve this? TCP achieves this by using a process called three way handshake. This process ensures that both sides are ready to communicate before data transfer begins. This setup process makes TCP reliable, but it also adds a bit of overhead, which we will be discussing later on.

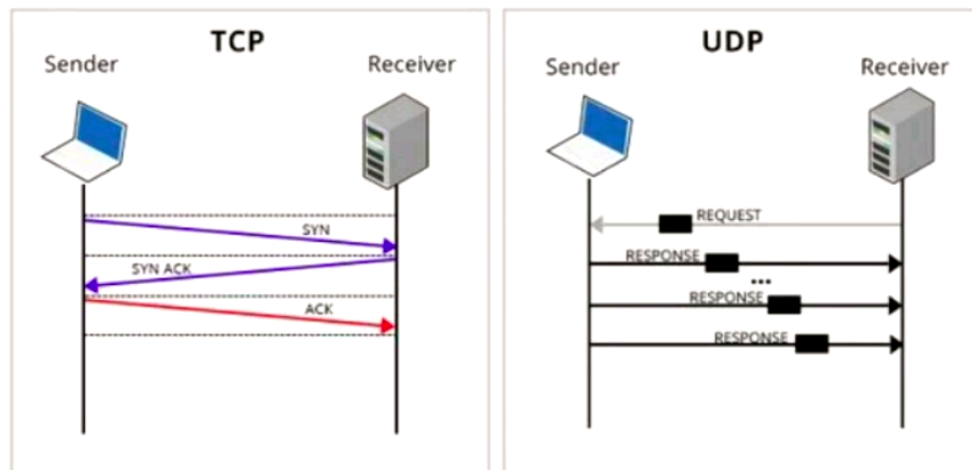
The biggest advantage of TCP is that it ensures reliable and ordered communication. Every piece of data sent via TCP is checked for errors. It is acknowledged by the receivers and reassembled in the correct order before being delivered. For example, imagine you are downloading a file or watching a video on YouTube, the data packets might take different routes to reach you. But TCP ensures that they all arrive and they are pieced together correctly in the current sequence so that you don't end up missing or scrambled content. If let's say any packet is lost or corrupted on the way, TCP will automatically request the sender to resend it, ensuring that no data is lost or delivered incorrectly.

Because of this reliability, TCP is used in applications where accuracy is critical. Think of scenarios like online banking, emails, web browsing or file transfers, where even a small piece of missing or incorrect data could cause major issues. The trade-off here is that TCP's error checking, retransmission and ordering mechanism adds some extra processing time, which makes it slightly slower than other protocols like UDP. But for applications where accuracy matters more than speed, TCP is the perfect choice.

So to summarize, TCP is a connection oriented protocol that ensures reliable, ordered and error free communication. It establishes a secure connection, it verifies data integrity and guarantees that the information is delivered correctly. This makes it the backbone of many critical internet applications.

What is User Datagram Protocol (UDP)?

While TCP focuses on reliability, UDP is designed for speed and efficiency. UDP is a very lightweight protocol that sacrifices some reliability related features to achieve lower latency. It is a connectionless protocol unlike TCP. This means that there is no need to establish a connection before sending data.



Imagine you are throwing a bunch of postcards into a mailbox without even waiting for confirmation that they are getting received or not. That's how UDP works. It simply sends data and does not check whether it was successfully delivered or not. Because UDP doesn't require a handshake or maintenance session, it significantly reduces the overhead and speeds up the communication. But we see what comes at the cost of reliability. So it is faster than TCP, but there are no delivery guarantees.

The biggest advantage of UDP is speed. Since it doesn't waste time on error checking, acknowledgement and retransmissions, it's much faster than TCP. This makes it ideal for real-time applications like online gaming, live streaming, voice chat, video call, where speed is more important than perfect accuracy. UDP doesn't guarantee that data will arrive at its destination, so packets can get lost in the transit. Packets might arrive out of order or even duplicated when they are reaching the destination. But there are applications that can tolerate minor data loss or they have their own error handling mechanism, where they can put these packets in sequence and that's where UDP is a better choice.

But since UDP doesn't provide delivery guarantees, lost packets are never retransmitted like they get retransmitted in the TCP. This means that if a data packet is dropped due to network congestion or any other issue, it is simply gone forever. Imagine you're watching a live sports stream, if one video frame is lost, the stream will keep on going and you wouldn't mind it. It doesn't wait to retrieve the missing frame because that could cause unnecessary lags in the stream. The same principle applies to online gaming, VIP calls, and the financial market updates, where receiving the latest data quickly is more important than getting the lost single packet.

So to summarize, UDP is connectionless. It's a high speed protocol that doesn't guarantee delivery. It's perfect for real-time applications where speed is more critical than perfect accuracy like gaming, video streaming, and voice communication. However, if you need reliable ordered data transmission, TCP is a better option.

Key Differences Between TCP & UDP

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Reliability	Reliable (ensures data delivery)	Unreliable (no guarantee of delivery)
Speed	Slower (due to error checking & retransmission)	Faster (no retransmission overhead)
Connection Type	Connection-oriented (establishes a connection before communication)	Connectionless (sends data without setup)
Ordering	Ensures packets arrive in order	No guarantee of packet order
Error Handling	Built-in error checking & retransmission	Minimal error checking, no retransmission
Overhead	High (due to handshaking, sequencing and acknowledgements)	Low (Minimal protocol overhead)
Use Cases	Web browsing (HTTP/HTTPS), File transfers (FTP, SFTP), Email (SMTP, IMAP, POP3), Database communication.	Video streaming (YouTube, Netflix), Online gaming, VoIP calls (Skype, Zoom), DNS lookups.

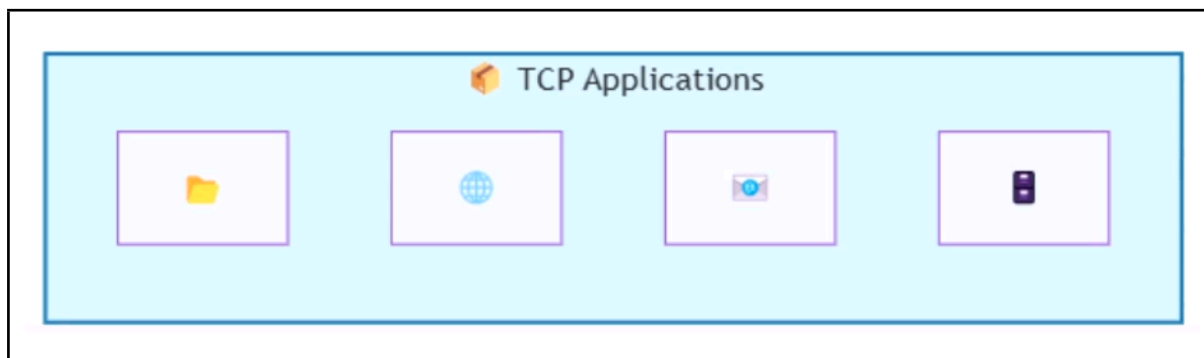
So to summarize, the key difference between these 2 protocols is that: TCP prioritizes reliability and accuracy, making it ideal for critical data transfers, whereas UDP sacrifices reliability for speed, making it perfect for speed, making it perfect for real-time applications.

Use Cases: When to use TCP vs. UDP

In this section, we are going to look at some of the real-world applications of these protocols. Choosing the right one will depend on whether you prioritize data integrity or speed.

- **When to use TCP?**

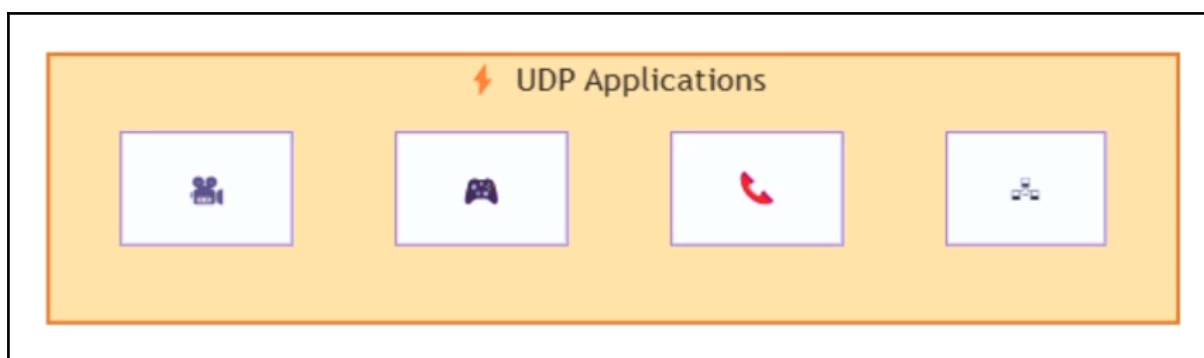
TCP is the preferred choice where accuracy and reliability are more important than speed. If a packet of data is lost or corrupted, TCP ensures it is retransmitted. This makes it perfect for applications where every bit of data must arrive correctly.



So web browsing, the protocols like HTTP and HTTPS are written on top of TCP. File transfer protocols like FTP and SFTP are on top of TCP. Digital communication, SMTP, IMAP, POP3, all these protocols are using TCP underlying.

- **When to use UDP?**

UDP is better when speed is more important than reliability. A few lost packets won't matter as much if the real-time responsiveness is maintained for the application.



Best use cases are video streaming, like youtube and netflix uses this. Online games like Fortnite, Call of Duty, PUBG are using it. VOIP calls, whatsapp calls, skype calls, all of them are using it. In fact DNS lookup also uses UDP underlying because when a website wants to resolve IP for a given domain name, it wants the response to be quick rather than late.

Summary & Key Takeaways

- TCP is reliable but slower; UDP is fast but unreliable.
- TCP is best for web pages, file transfers, and email.
- UDP is best for real-time applications like video calls and gaming.
- System design must balance speed, reliability, and user experience.
- What's Next:
 - HTTP - The Backbone of the Web

Interview Questions: TCP & UDP

- Basic Questions:
 - What is the difference between TCP and UDP?
 - Why is TCP considered a reliable protocol?
 - When would you choose UDP over TCP?
- Technical & Deep-Dive:
 - How does TCP ensure reliable data transmission?
 - What is the Three-Way Handshake in TCP?
 - How does UDP handle packet loss?
- Use Case & System Design:
 - Which protocol would you use for a real-time multiplayer game? Why?
 - How does TCP handle congestion control?
 - Why is DNS implemented over UDP instead of TCP?
 - If a video streaming service is buffering too much, would switching from TCP to UDP help?

Pro Tip: Be prepared to compare TCP & UDP in real-world scenarios and discuss trade-offs in speed vs. reliability!